

Why Separate the Roles?

Clinical prediction from EHRs needs locally deployable models when closed-model APIs raise privacy and compliance concerns. Most medical multi-agent systems are evaluated as one unit, obscuring which role changes the prediction.

Q1. Which agent role changes the point at which a patient is flagged?

Q2. Can open-weight agents approach closed-model prompting without task-specific training?

Controlled Role Design

Two contrastive analysts

Risk: compare the target EHR with a positive-outcome exemplar.

Protective: compare it with a negative-outcome exemplar. Neither analyst sees the target label or makes the final prediction.

One integrator

Reconcile both readings, retrieve additional medical knowledge, and output risk $\hat{p}$; flag when $\hat{p} > 0.5$.

Studied assignment

GPT-OSS-120B analysts $\rightarrow$ Qwen2.5-7B integrator

MedCorp + UMLS knowledge and the two patient exemplars are fixed across role-assignment comparisons.

Four Evaluation Settings

Positive-outcome prevalence (about 1,000 test EHRs per setting)

Outcome	MIMIC-III	MIMIC-IV
In-hospital mortality	5.4%	19.2%
15-day readmission	54.8%	46.5%

Why report sensitivity? On MIMIC-IV mortality, predicting every patient as a survivor gives **80.8% accuracy** but **0% sensitivity**.

Pipeline: Fixed Retrieval and Specialized Roles

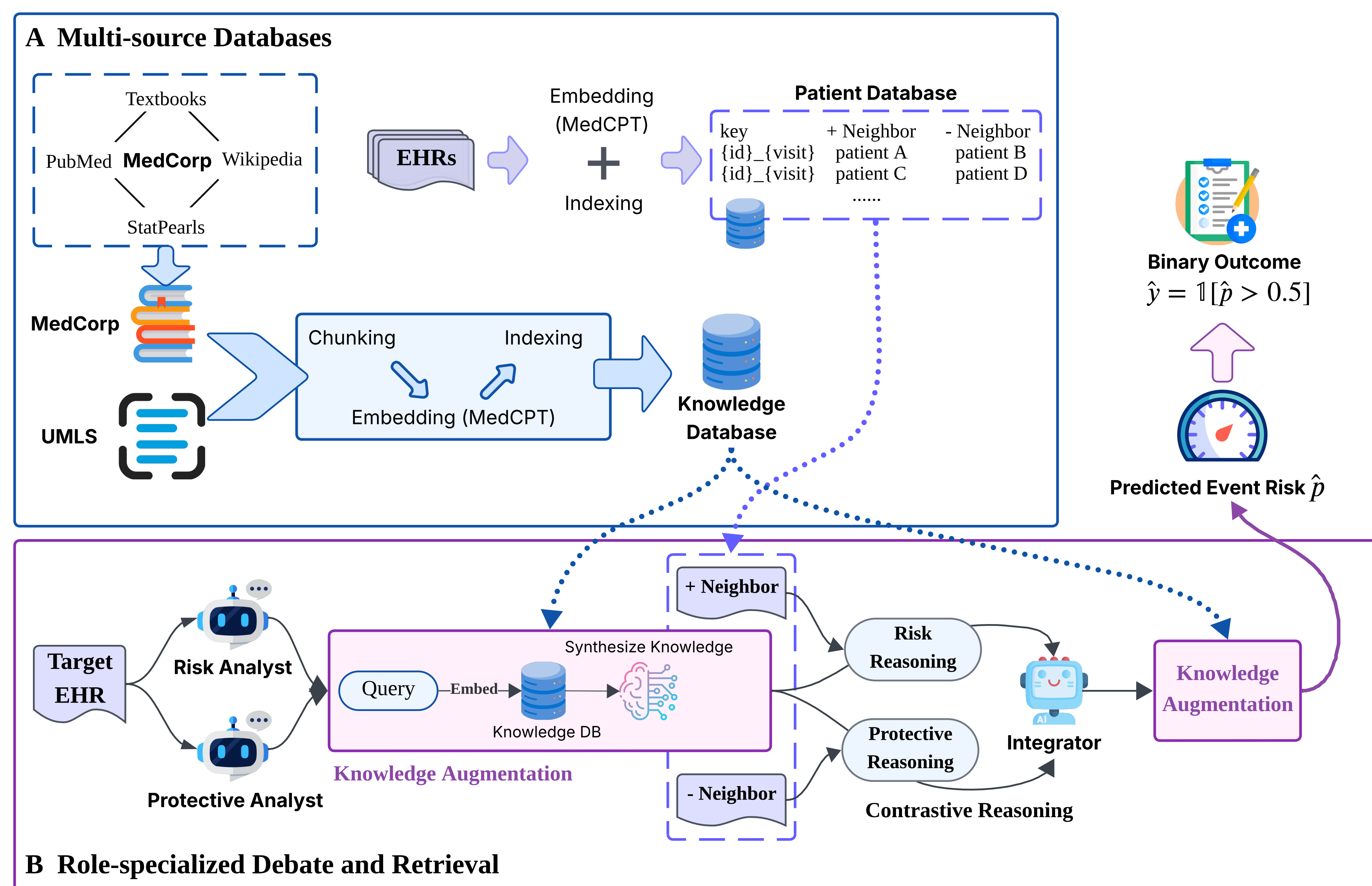


Figure 1. Medical knowledge and similar patients support two contrastive readings. The integrator combines them, runs its own focused retrieval, and produces the final risk score.

Q1: Which Role Changes Sensitivity?

MIMIC-IV mortality, role-isolation subset ($n = 100$)

Sensitivity at the 0.5 cutoff (%)

All GPT-OSS-120B	13.0
Only retrieval model $\rightarrow$ Qwen	7.4
Only analyst $\rightarrow$ Qwen	11.1
Only integrator $\rightarrow$ Qwen	64.8

Changing only the integrator

Macro-F1: 38.5 $\rightarrow$ 57.4 AUROC: .543 $\rightarrow$.603

The analyst or retrieval-model swap does not produce this high-recall operating point. These are **subset results**, separate from the full-test results at center.

Component Check: Retrieval and Roles

GPT-OSS-120B, MIMIC-IV mortality; macro-F1 (%)

Single, no retrieval	58.3
Single + medical RAG	37.7
Single + RAG + exemplars	52.8
Three roles + RAG + exemplars	54.4

With the same retrieved inputs, three roles raise macro-F1 from **52.8 to 54.4**. The no-retrieval single model still scores **58.3**; retrieval alone is not the source of the main result.

Q2: Mortality on the Full Test Set

MIMIC-IV, about 1,000 patients; 19.2% positive; threshold 0.5

Claude 3.5 few-shot Mixed open-weight system

Metric	Claude 3.5 few-shot	Mixed open-weight system
Macro-F1 (%)	57.4	58.8
Sensitivity (%)	15.8	61.1
Accuracy (%)	84.5	66.4

More deaths flagged, more false positives. Macro-F1 is similar to Claude few-shot, while sensitivity rises by **45.3 points** and accuracy falls by **18.1 points**.

The mixed system's full-test AUROC is **.706**, versus **.763** for a single GPT-OSS agent. The benefit is at the default cutoff, rather than stronger ranking. Task-supervised KARE has higher macro-F1 (90.4).

Integrator Swap Across All Test Sets

Qwen instead of GPT-OSS at integration; analysts and retrieval model fixed

Outcome	Δ Sens.	Δ Macro-F1
MIMIC-IV mortality	+46.4	+4.4
MIMIC-III mortality	+37.1	-16.7
MIMIC-IV readmission	+51.0	-12.5
MIMIC-III readmission	+54.1	-12.6

Changes are **percentage points** versus the all-large pipeline. Sensitivity rises in all four settings, but macro-F1 improves only on MIMIC-IV mortality.

Mechanism: Where Risk Crosses 0.5

MIMIC-IV mortality (survivors)

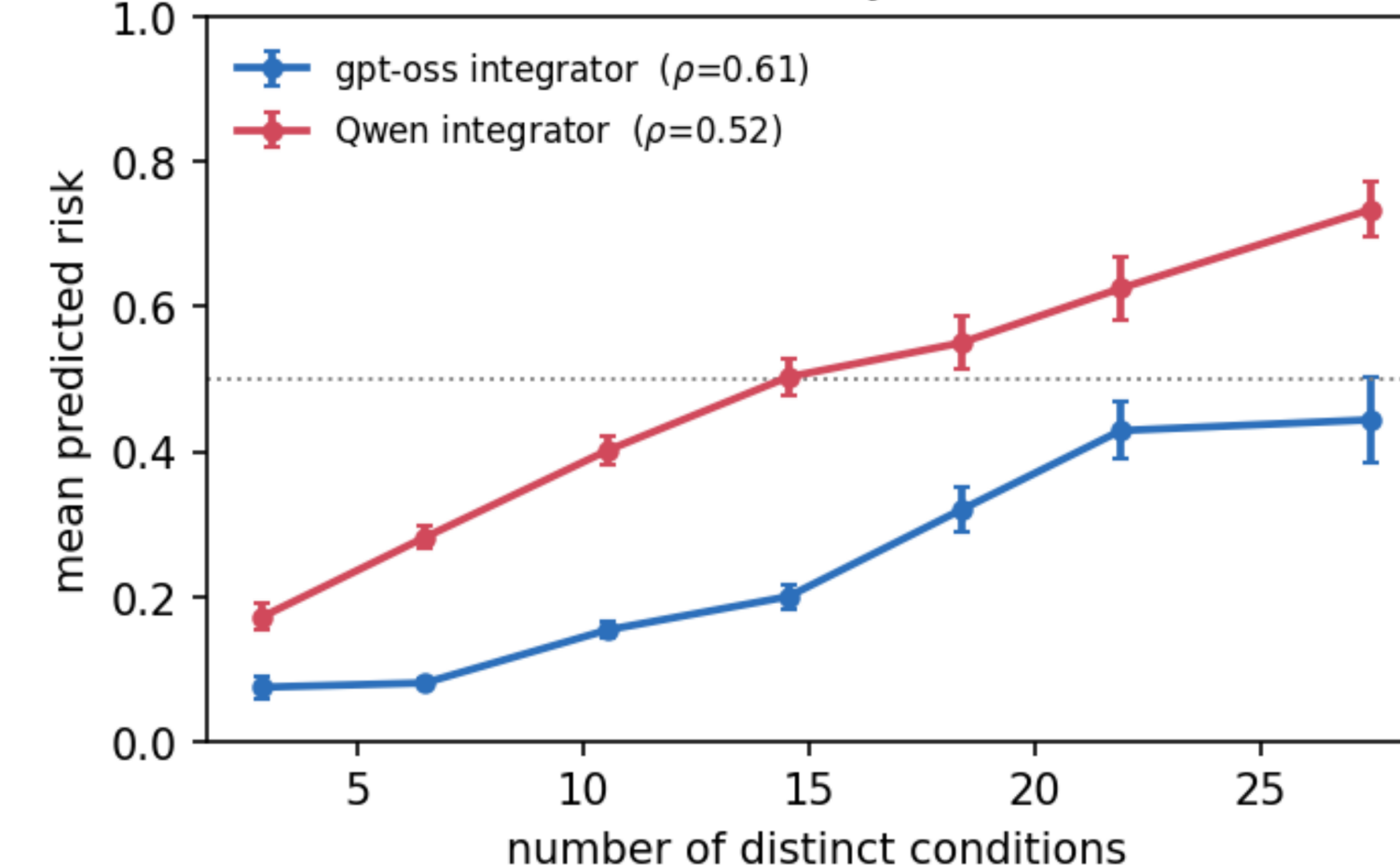


Figure 2. Among *survivors*, both integrators raise predicted risk as recorded conditions increase. Qwen crosses the 0.5 decision threshold earlier, creating more false positives. Error bars show standard error. The mixed system also flags **56%** of decedents missed by the all-large system.

Boundary: Readmission

MIMIC-IV, all-large $\rightarrow$ mixed

Sensitivity 43.5 $\rightarrow$ 94.5 Macro-F1 47.2 $\rightarrow$ 34.7

The lower threshold gives about **4% specificity**. A bag-of-codes probe reaches AUROC **.77** for mortality but only **.59** for readmission on MIMIC-IV: the available record has weaker signal for this post-discharge outcome.

Scope. Evaluated on MIMIC-III/IV mortality and 15-day readmission, using two open-weight model families. No external or prospective validation; the result does not surpass fine-tuned models on overall macro-F1.



Paper



Code